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MULTI-PATIENT EVALUATION OF COMPUTATIONAL MODELS OF UTERINE MECHANICS

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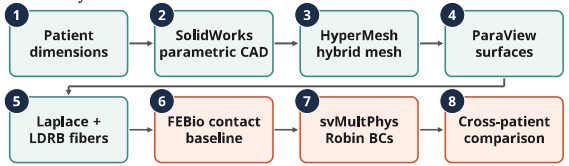
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UNMET CLINICAL NEEDS

- Only ~5% of pregnancies deliver on the estimated due date, and **preterm birth (<37 weeks) affects ~10% of births** worldwide [1]
- **Image-based finite element models** can noninvasively probe the mechanical environment of the uterus and cervix that regulates labor onset [1]
- A cardiac-mechanics-inspired pipeline matched a contact-based FEBio model with average nodal displacement differences of **3.1 ± 1.6 mm at ~90% lower cost**, but was **developed by Mergler et al. on a single patient geometry** [1]

STUDY AIM: ONE PATIENT → FIVE

Extend and streamline the single-patient pipeline [1] to five patient-specific geometries (P1–P5), testing whether one standardized workflow, and one set of tuned parameters, transfers across anatomical variability



*Repeated for **all five patients**; teal = geometry & model setup, orange = simulation & analysis*

PATIENT-SPECIFIC GEOMETRY

- Parametric **SolidWorks** template developed by Louwagie et al. driven by a design table of sagittal, axial, and cervical measurements from ultrasound/MRI [2, 3]
- Built **four new patient models (P2–P5)** and compared them against the original reference models in ParaView

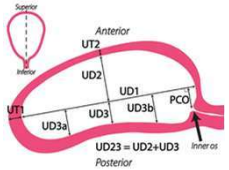
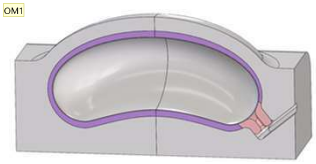


Image-based measurements [2]

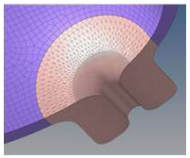


Patient SolidWorks assembly: uterus, cervix, membrane, abdomen

HYBRID MESHING

- Meshed every component in **HyperMesh** with **continuous nodes** across uterus–cervix and model halves (Boolean with shared interface), so no contact is needed between tissues [1]

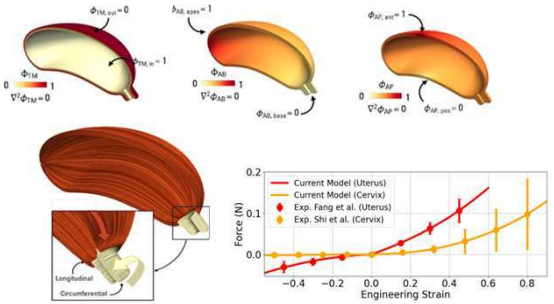
Component	Elements	Size
Uterus	HEX8 solid map, quads only (≥3 through thickness)	2 mm
Cervix	TET4, surface-seeded	0.4–2 mm
Fetal membrane	HEX8 solid map (1 through thickness)	2 mm
Abdomen	TET4, graded outward	2–10 mm



Shared nodes: HEX8 / TET4

LAPLACE FIELDS, FIBERS, & MATERIAL

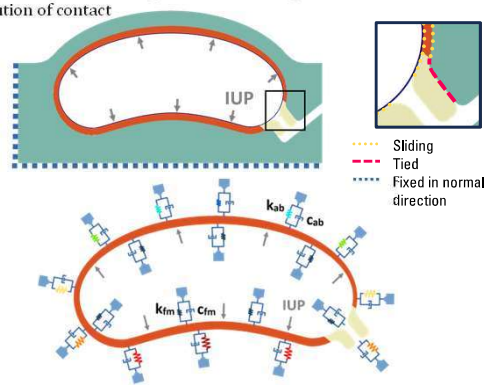
- Extracted labeled **surfaces** for each patient in ParaView with global node and element IDs for svMultPhys
- Solved **four Laplace (heat-transfer) problems** per patient in FEBio on **SDSC Expanse**: uterine and cervical apico-basal and transmural fields
- **LDRB fibers** [4]: longitudinal in the uterus, circumferential in the cervix, blended through the isthmus [1]
- **GOH mechanical model** used to characterize uterus and cervix material properties with parameters tuned to experimental data of the tissues in tension and compression [1]



Laplace-Dirichlet fields and resulting LDRB fiber directions (P1 shown). GOH material model fit against experimental data. [1]

BOUNDARY CONDITIONS & SOLVERS

- **Verification baseline (FEBio) configuration for all five patients**
  - Employed methods developed in Louwagie et al. [3]
  - Translated fiber architecture to FEBio model
  - Configured contact with abdomen and fetal membrane
  - Combinations of sliding and tied contacts to capture physiological interactions
  - Applied physiological intrauterine pressure to inner surface of fetal membrane
- **New model (svMultPhys) configuration for all five patients**
  - Using previously developed pipeline where laplace solutions are used to configure spatially varying **Robin Boundary Conditions**
  - Replace the surrounding tissue, eliminating the configuration and solution of contact



Comparison of contact-based verification data and current model boundary condition configuration protocols. [1]

RESULTS: CROSS-PATIENT COMPARISON

5	4	10	3.6 mm
patient geometries	new models (P2–P5)	full simulations (5 FEBio + 5 svMultPhys)	mean nodal difference (pooled)

Nodal displacement difference: FEBio contact vs. svMultPhys Robin BC



All patients (580,749 nodes): Average: 3.6 ± 2.2 mm **Max:** 12.7 mm

- Mean nodal displacement differences of **2.8–5.7 mm** across patients (pooled **3.6 ± 2.2 mm**)
- Robin BC parameters tuned on P5 [1] transferred **without re-tuning**: P2–P4 means were within 0.4 mm of P5; **P1 was the outlier** (5.7 ± 3.3 mm)
- Largest differences at the **lateral wall** (coronal) and **posterior wall** (sagittal); the **cervix agreed most closely**, as in the single-patient study [1]

CONCLUSIONS & FUTURE WORK

- Scaled the pipeline from **one to five patient geometries** with a single documented protocol spanning geometry, meshing, surfaces, fibers, and solvers
- Resolved scaling bottlenecks: fillet-driven stress concentrations, node continuity, mesh conversion for svMultPhys
- Quantified per-patient nodal displacement agreement (**2.8–5.7 mm** mean difference): **single-patient BC parameters generalized to P2–P4 without re-tuning**, while P1 showed the largest differences
- **Long term:** automated, cohort-scale models for longitudinal growth & remodeling and contraction studies

ACKNOWLEDGEMENTS AND REFERENCES

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[1] Mergler, O., et al. A method for image-based modeling of uterine passive mechanics during late pregnancy. [2] Louwagie, E.M., et al., 2021. PLoS ONE 16(1): e0242118. [3] Louwagie, E.M., et al. Parametric solid models of the at-term uterus from 2D ultrasound and MRI. Eng. Comput. [4] Bayer, J.D., et al., 2012. Ann. Biomed. Eng. [5] Strocchi, M., et al., 2020. J. Biomech.